World Bank Rule of Law Outputs

This repository now uses the official World Bank Worldwide Governance Indicators (WGI) rule-of-law series `RL.EST` (`Rule of Law: Estimate`) as the rule-of-law variable for all current World Bank-based figures and merged regression outputs.

Scope

- Main governance variable: `RL.EST`
- Main analysis window: `1996-2015`
- Requested display window for the regional comparison figure: `1990-2015`
- Important data note: official `RL.EST` values begin in `1996` , so `1990-1995` appear as blank years where relevant

Indicator Definitions

1. `RL.EST`

Item	Definition
Code	`RL.EST`
Name	Rule of Law: Estimate
Source	World Bank, Worldwide Governance Indicators (WGI)
Meaning	Perception-based estimate of confidence in and compliance with the rules of society
Core dimensions	Contract enforcement, property rights, police, courts, crime and violence
Scale	Standardized governance estimate, typically interpreted on an approximate `-2.5` to `+2.5` scale

Interpretation:

- Higher values indicate stronger perceived rule of law.
- Lower values indicate weaker perceived rule of law.
- `0` is roughly around the middle of the WGI comparison scale.
- The values are not percentages.

- The values are not spending levels, grant amounts, or project-intensity scores.

2. `NY.GDP.MKTP.KD.ZG`

Item	Definition
Code	`NY.GDP.MKTP.KD.ZG`
Name	GDP growth (annual %)
Source	World Bank
Use in this project	Dependent variable in the main scatter and regression outputs

3. `NY.GDP.MKTP.KD`

Item	Definition
Code	`NY.GDP.MKTP.KD`
Name	GDP, constant 2015 US$
Source	World Bank
Use in this project	Level control; transformed into `log_gdp` in the regression workflow

4. `SP.DYN.LE00.IN`

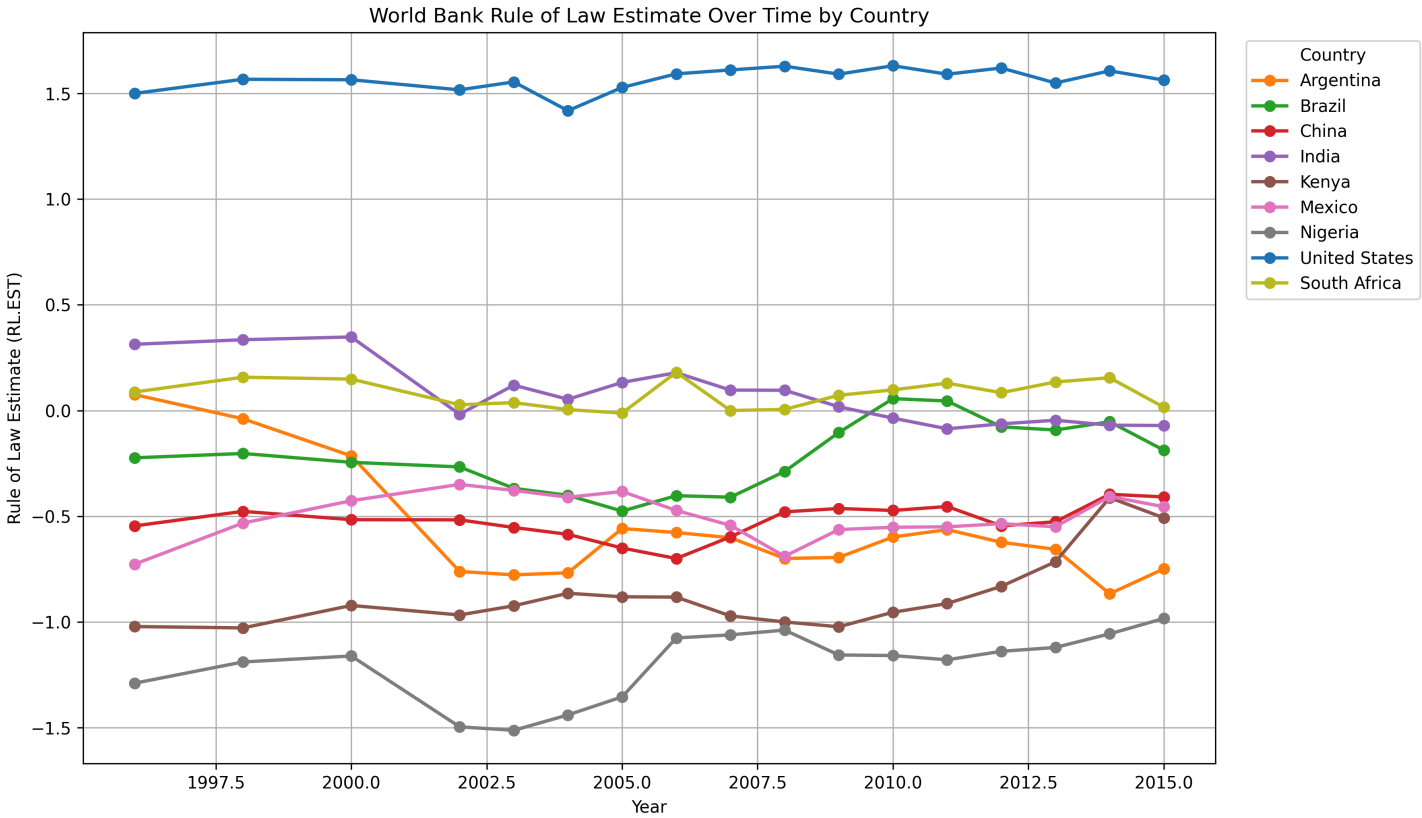
Item	Definition
Code	`SP.DYN.LE00.IN`
Name	Life expectancy at birth, total (years)
Source	World Bank
Use in this project	Additional control variable in the regression workflow

Data Files Used

- `CSV_file/WB_index/API_RL.EST_DS2_en_csv_v2_5814.csv`
- `merged_gdp_rule_of_law_dataset.csv`
- `period_averages_gdp_rule_of_law.csv`
- `rl_est_us_china_japan_africa_latam_euro_area_1990_2015.csv`

Figure Guide

1. Combined Rule-of-Law Trend by Country



File:

- `rule_of_law_estimate_over_time_by_country.png`

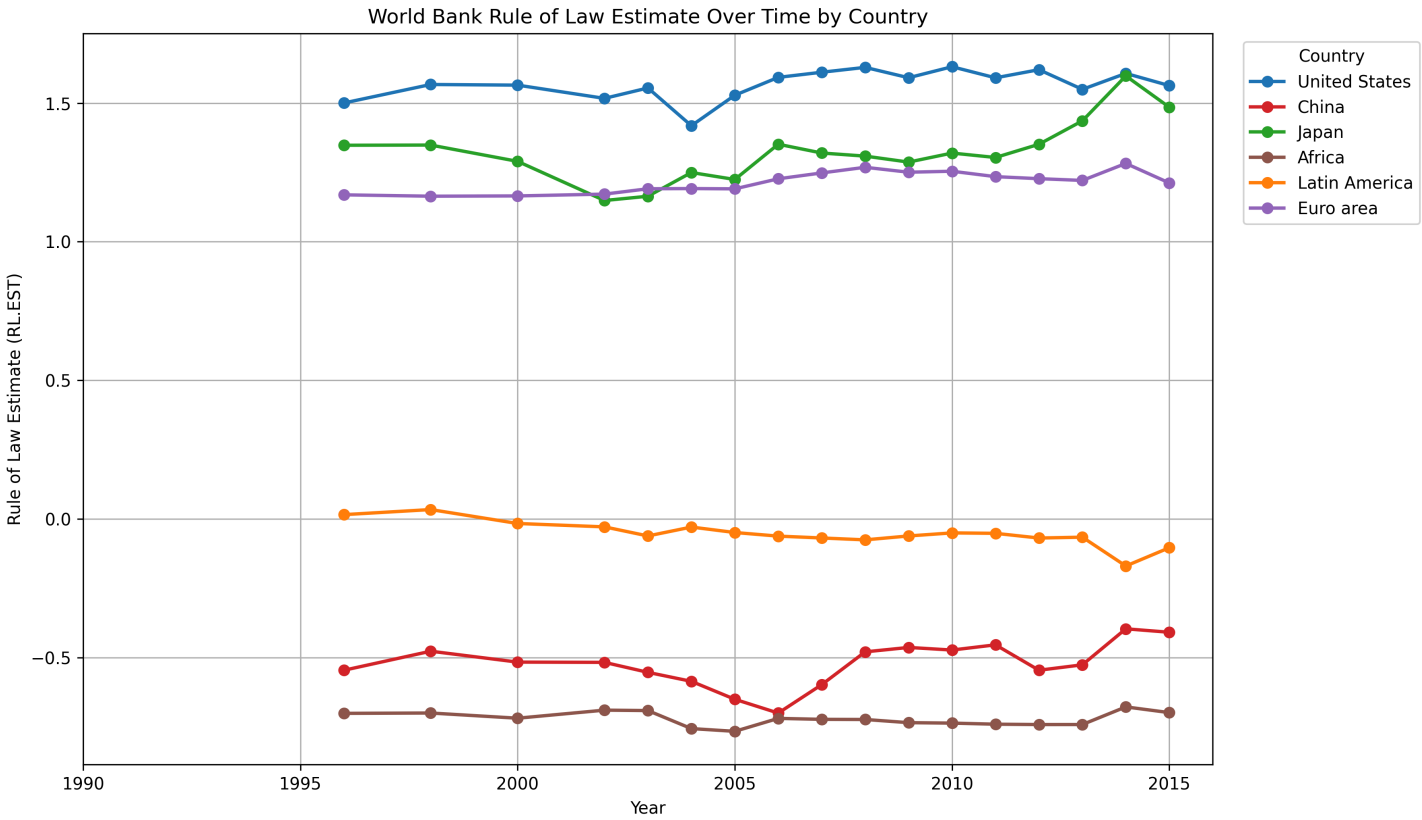
What it shows:

- The official World Bank `RL.EST` series for the project country sample
- Cross-country comparison of long-run rule-of-law trajectories
- The relative position of each country on the WGI rule-of-law scale

How to read it:

- Positive values indicate above-average perceived rule of law.
- Negative values indicate below-average perceived rule of law.
- Year-to-year changes should be interpreted cautiously when they are small.

2. Regional Comparison Figure



Files:

- `rl_est_us_china_japan_africa_latam_euro_area_1990_2015.png`
- `rl_est_us_china_japan_africa_latam_euro_area_1990_2015.csv`

What it shows:

- `RL.EST` for the United States, China, and Japan
- Regional comparison series for Africa, Latin America, and the Euro area
- A unified view of advanced-economy, large-country, and regional trajectories

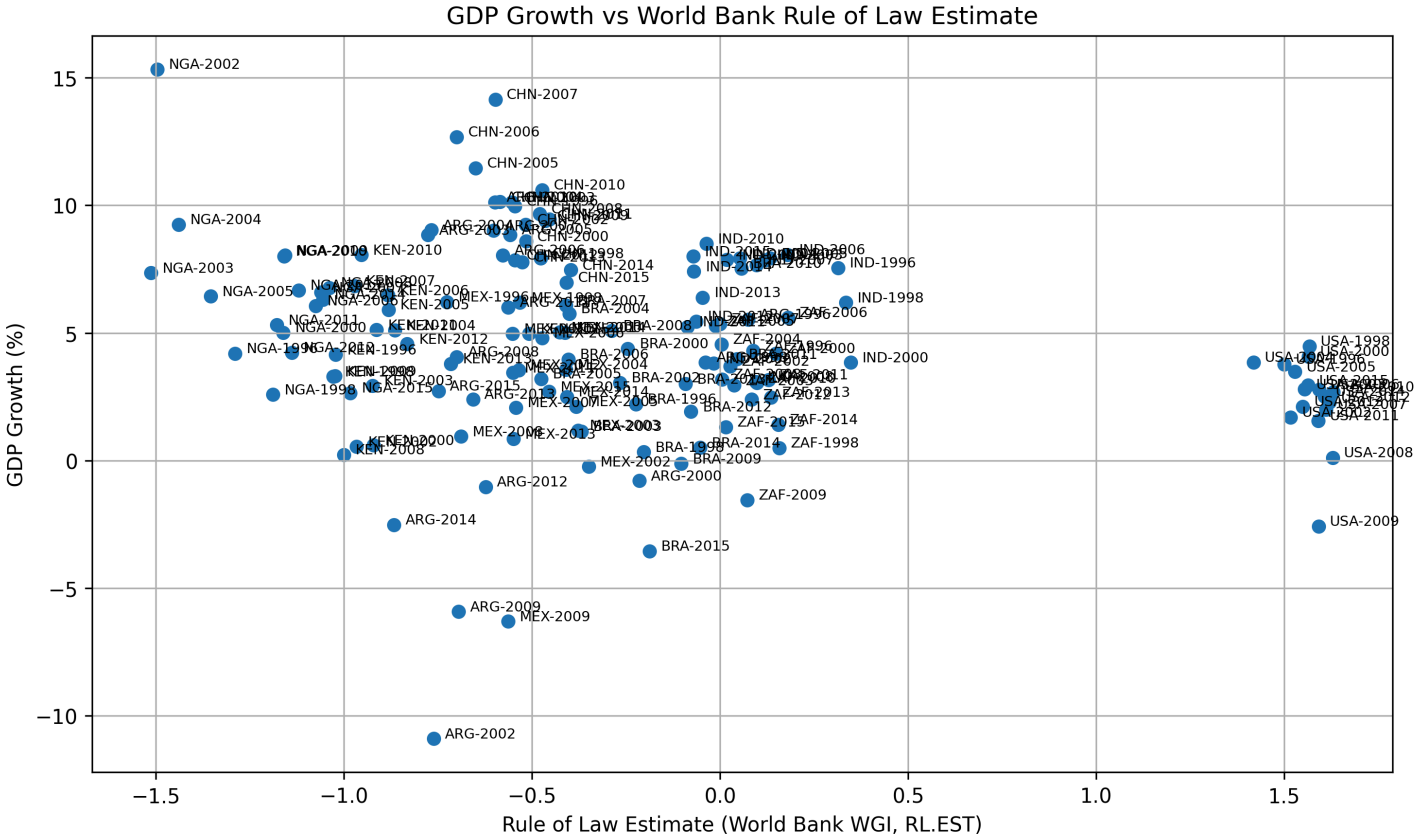
Regional construction:

- `Africa` is the simple annual mean of countries tagged as `Sub-Saharan Africa` in the World Bank country metadata.
- `Latin America` is the simple annual mean of countries tagged as `Latin America & Caribbean` .
- `Euro area` is the simple annual mean of 2015 euro-area member states used in the script.

Important note:

- Although the chart spans `1990-2015` , all six series begin in `1996` because `RL.EST` is not available before 1996.

3. GDP Growth vs Rule of Law Scatter



File:

- `gdp_vs_rule_of_law_scatter.png`

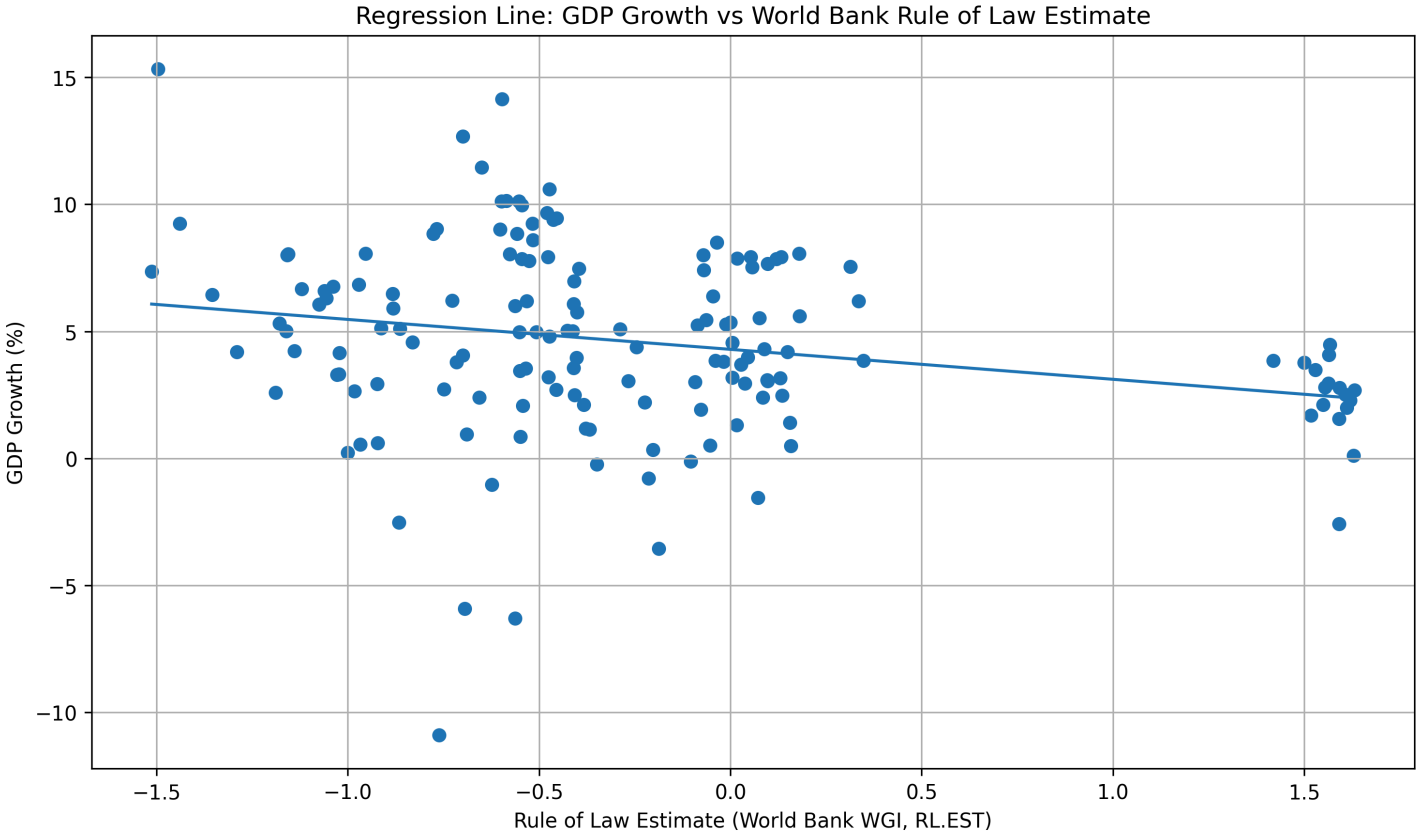
What it shows:

- Country-year observations with `RL.EST` on the horizontal axis
- GDP growth on the vertical axis
- The raw cross-sectional relationship in the selected sample

Reading:

- This figure is descriptive, not causal.
- The sample does not show a strong positive unconditional relationship between higher `RL.EST` and higher contemporaneous GDP growth.

4. GDP Growth vs Rule of Law Regression Line



File:

- `gdp_vs_rule_of_law_regression_line.png`

What it shows:

- The fitted linear relationship between GDP growth and `RL.EST`
- A visual summary of the simple OLS pattern in the sample

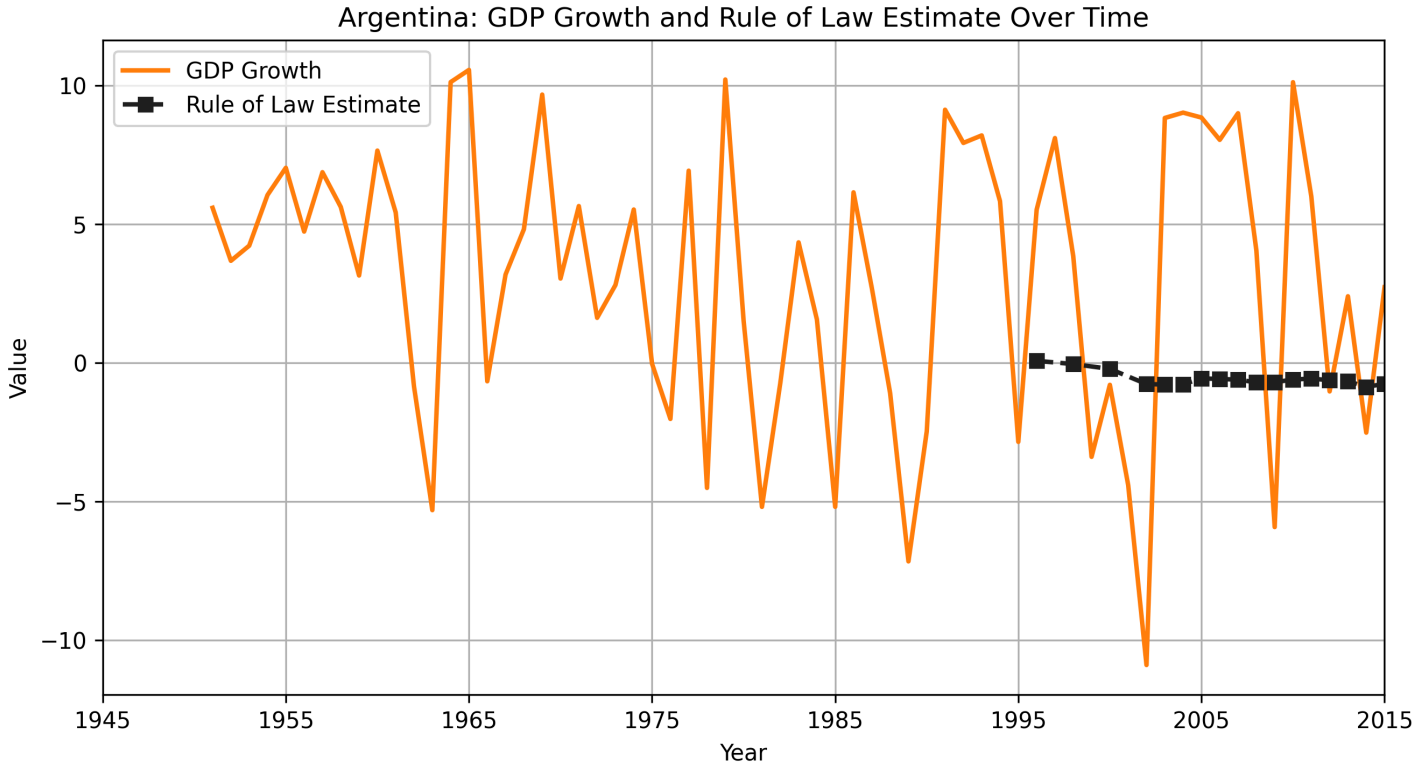
Reading:

- The fitted slope is negative in the current sample.
- This should be read as a sample pattern, not as proof that stronger rule of law reduces growth.

5. Country Trend Panels

Each country panel combines GDP growth and the World Bank rule-of-law estimate over time.

Argentina



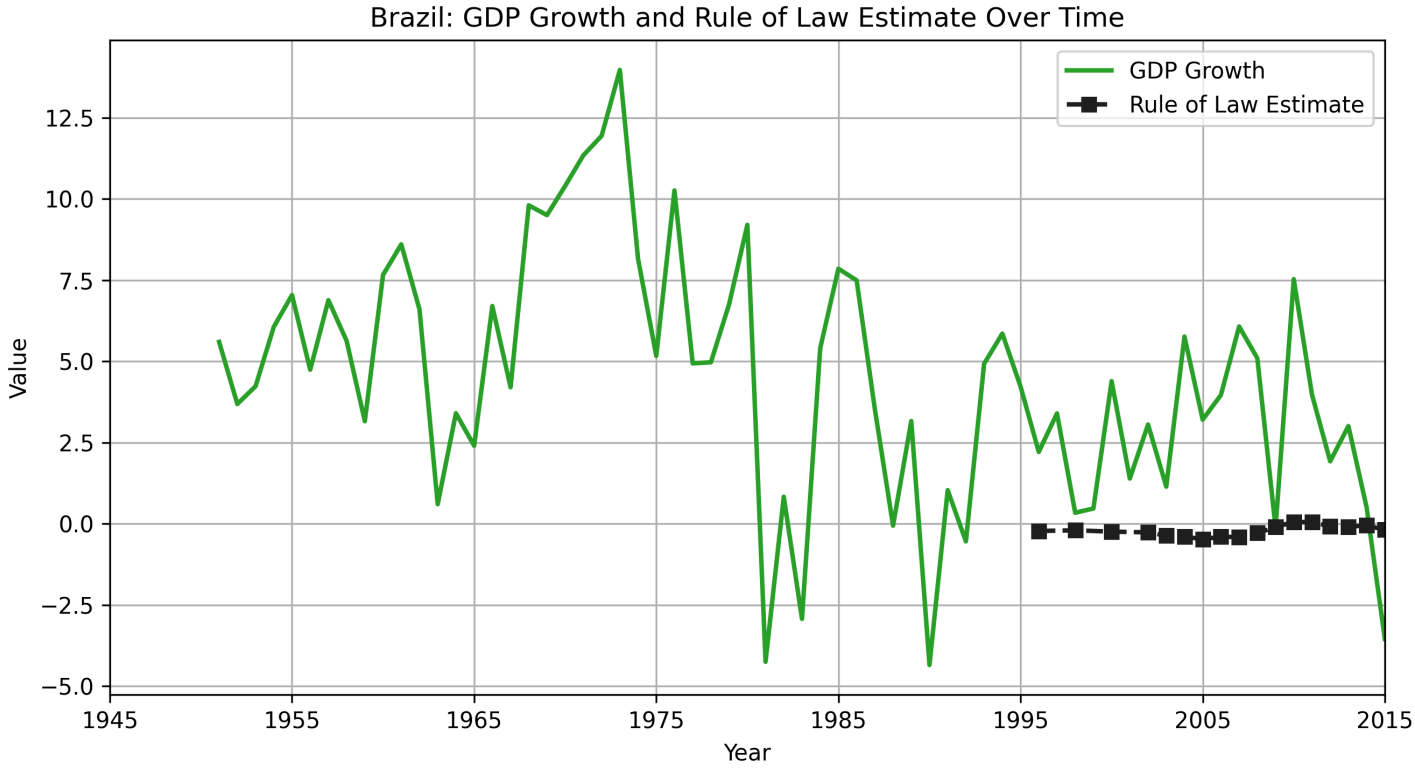
File:

- `ARG_trend.png`

Reading:

- Argentina trends from slightly positive rule-of-law values in the late 1990s to clearly negative values by 2015.

Brazil



File:

- `BRA_trend.png`

Reading:

- Brazil stays close to the midpoint of the WGI scale and fluctuates over time.

China

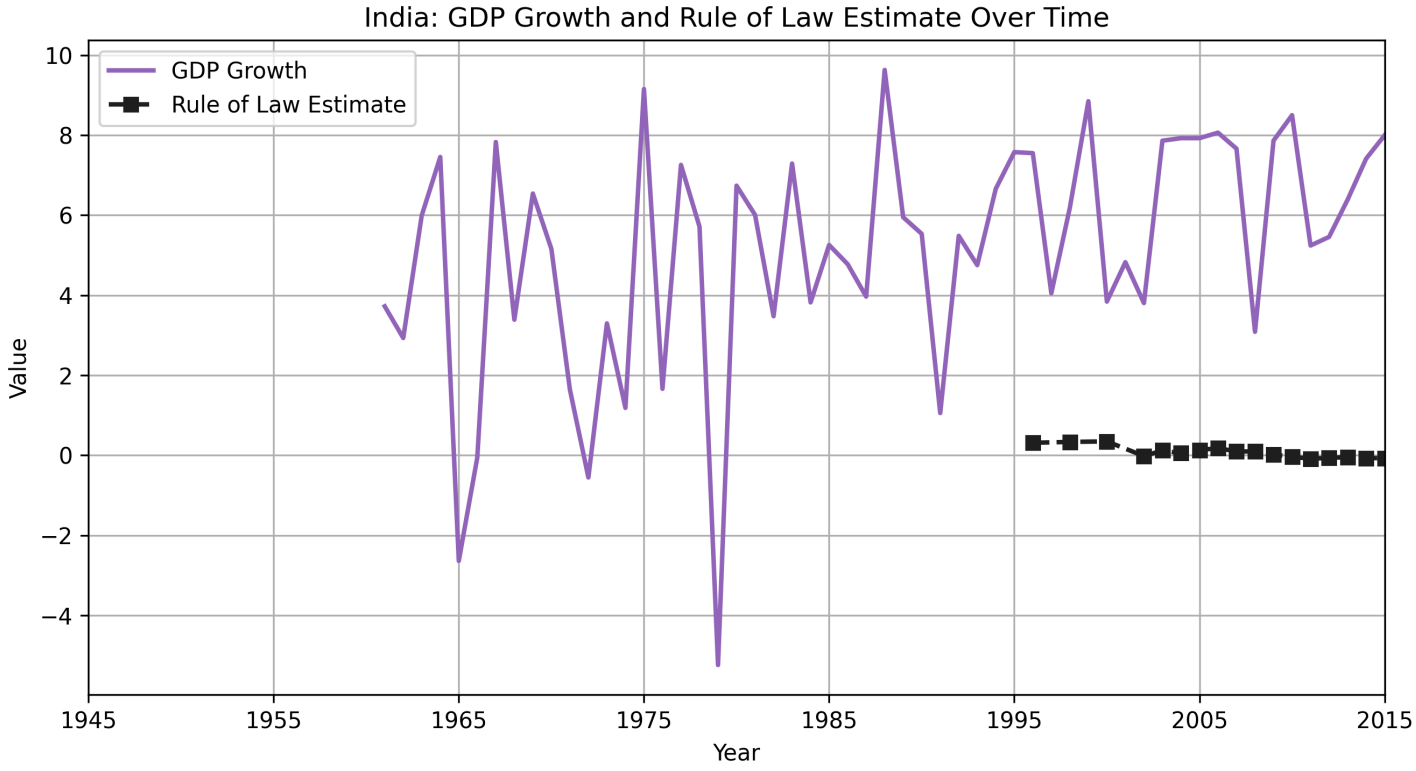
File:

- `CHN_trend.png`

Reading:

- China remains below zero throughout the sample but improves modestly relative to its starting point.

India



File:

- `IND_trend.png`

Reading:

- India starts positive and trends downward toward the midpoint by the 2010s.

Kenya

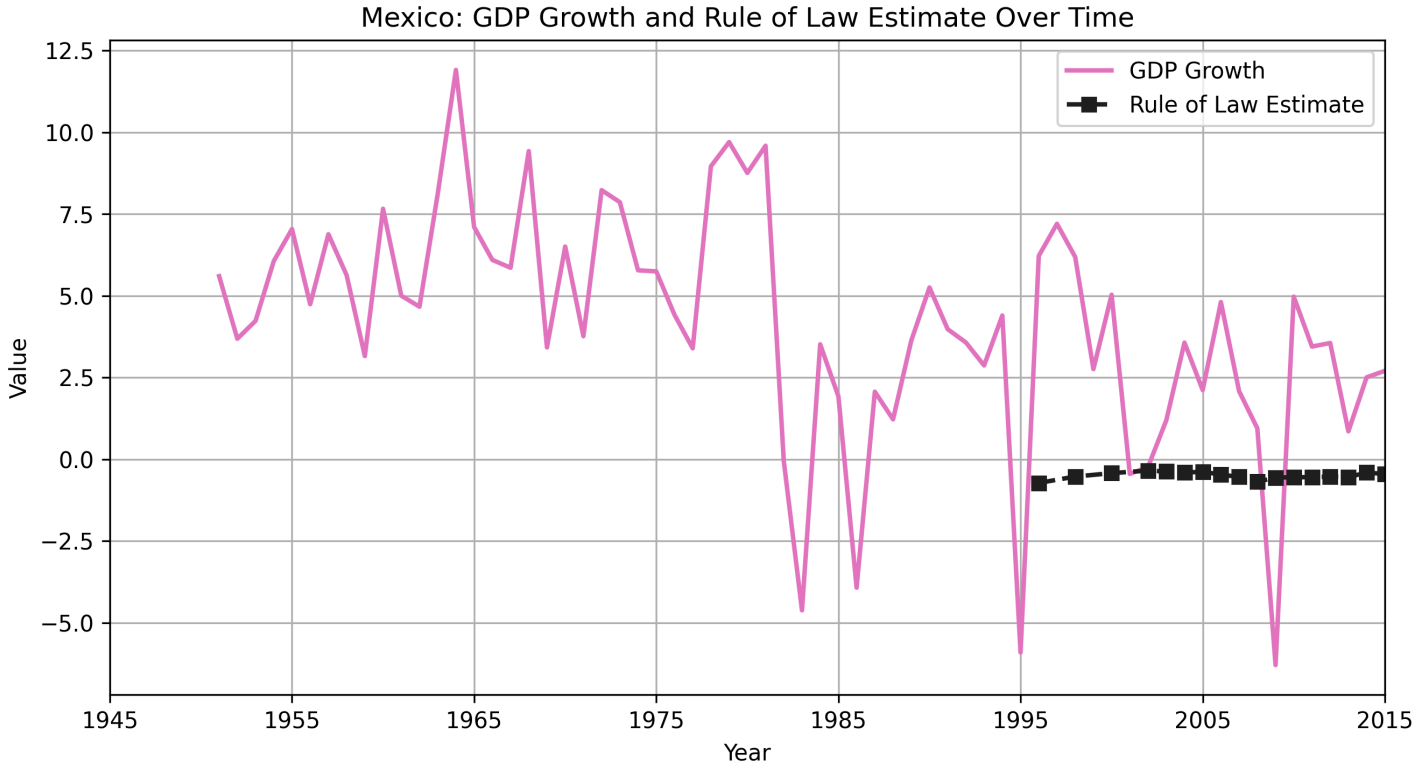
File:

- `KEN_trend.png`

Reading:

- Kenya remains below average on `RL.EST` but improves materially over the period.

Mexico



File:

- `MEX_trend.png`

Reading:

- Mexico remains negative throughout the sample with moderate long-run improvement.

Nigeria

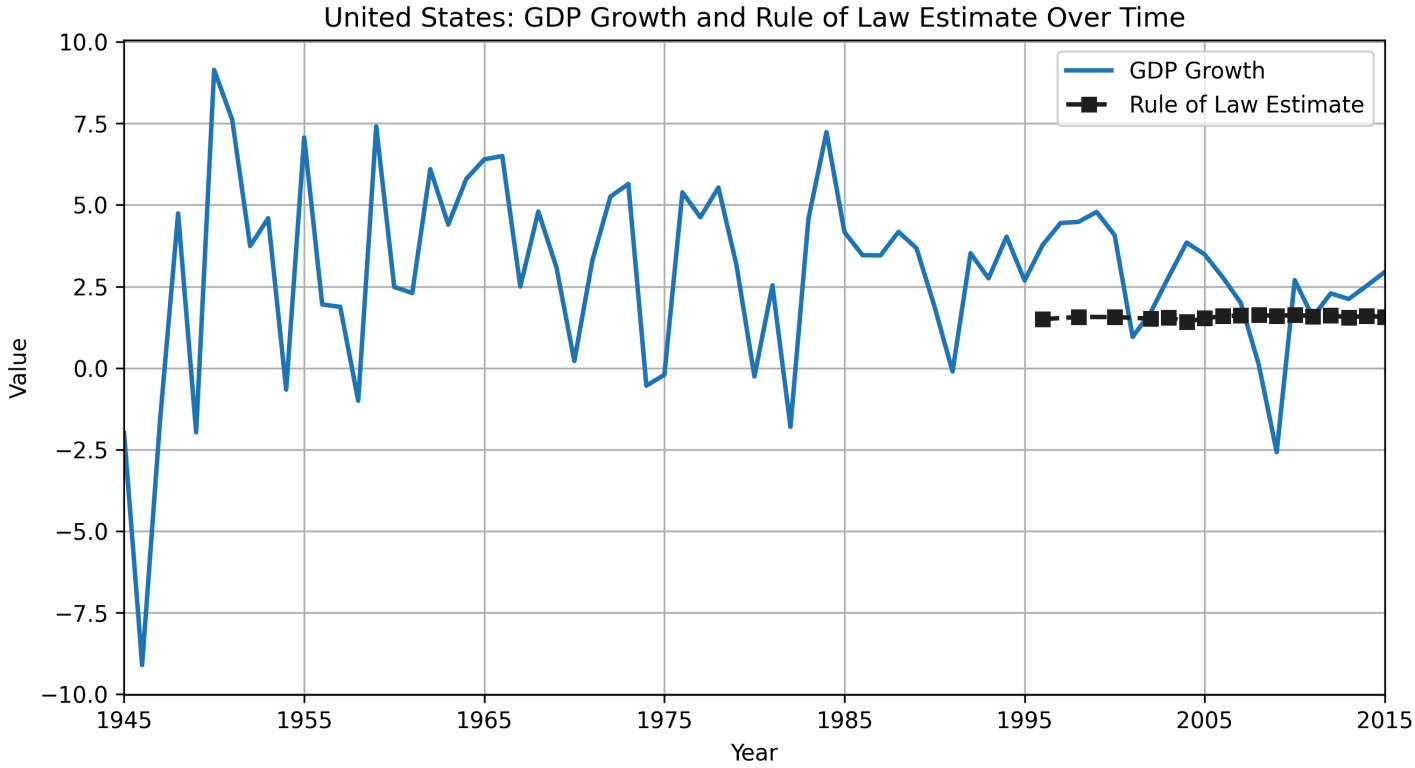
File:

- `NGA_trend.png`

Reading:

- Nigeria is persistently weak on `RL.EST`, though it improves somewhat by 2015.

United States



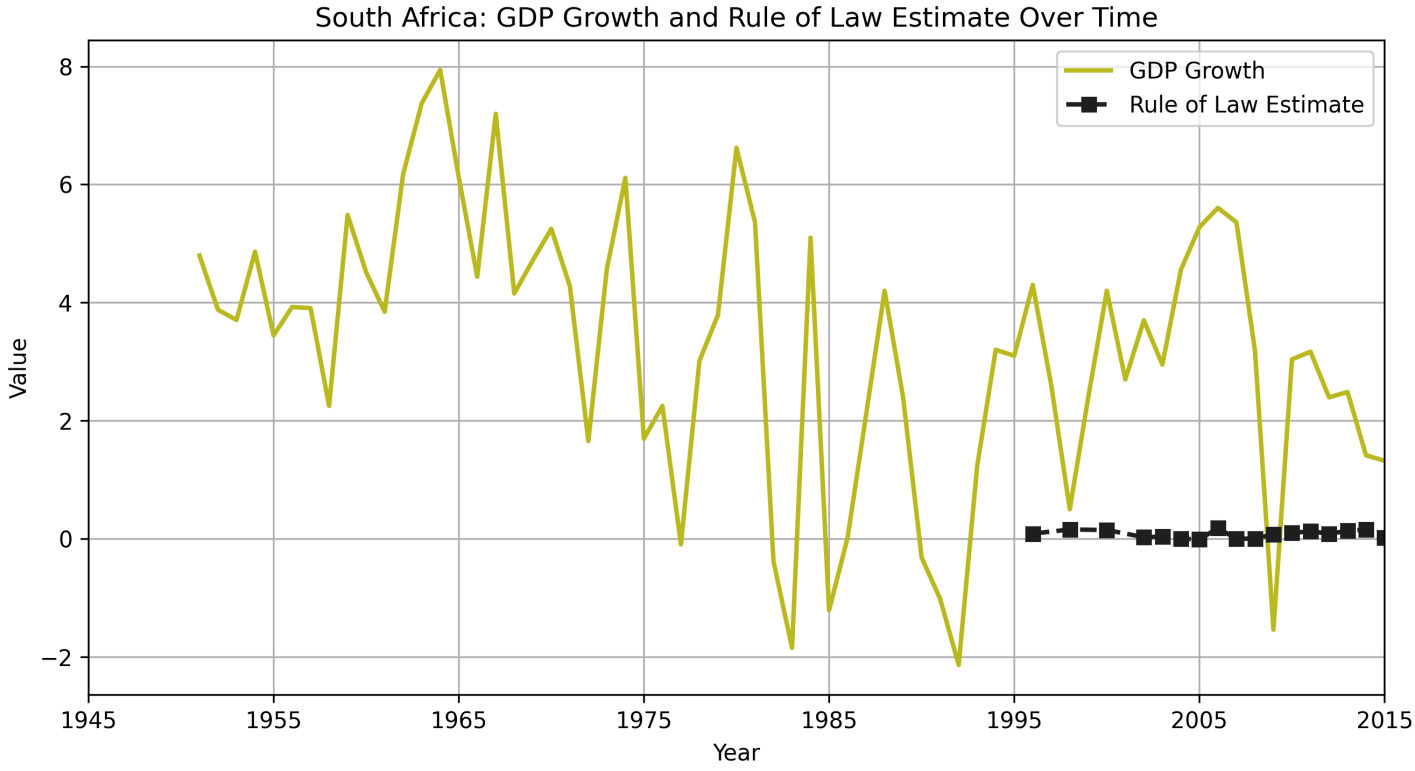
File:

- `USA_trend.png`

Reading:

- The United States remains consistently high and positive across the sample.

South Africa



File:

- `ZAF_trend.png`

Reading:

- South Africa stays close to the midpoint, with fluctuation rather than a strong directional trend.

Regression Summary

The current World Bank-based regression outputs report the following headline results:

Model	Coefficient on `RL.EST`	p-value	R-squared	Interpretation
Simple OLS	-1.178	0.0001	0.059	Negative cross-country association, low explanatory power
OLS with controls (`log_gdp` , `life_expectancy`)	-2.481	<0.0001	0.220	Negative association remains after controls
Country fixed effects	-1.176	0.564	0.405	Within-country effect is not statistically

Model	Coefficient on `RL.EST`	p-value	R-squared	Interpretation
				significant
Period regression `1996-2005`	-0.919	0.0648	0.037	Negative, marginal significance
Period regression `2006-2015`	-1.368	0.0001	0.079	Stronger negative association in the later period

Main takeaway:

- The negative sign is driven mainly by cross-country differences in the selected sample.
- Once country fixed effects are included, the rule-of-law coefficient is not statistically significant.

Interpretation Rules

- Describe the rule-of-law variable as the official World Bank WGI `RL.EST` series.
- Do not describe `RL.EST` as a funding index.
- Do not describe `RL.EST` as a Ford Foundation program score.
- Do not interpret the values as percentages.
- Do interpret `RL.EST` as a perception-based latent governance estimate.

Current Canonical World Bank Figure Set

The repository root now serves as the canonical location for the current World Bank output figures:

- `rule_of_law_estimate_over_time_by_country.png`
- `rl_est_us_china_japan_africa_latam_euro_area_1990_2015.png`
- `gdp_vs_rule_of_law_scatter.png`
- `gdp_vs_rule_of_law_regression_line.png`
- `ARG_trend.png`
- `BRA_trend.png`
- `CHN_trend.png`
- `IND_trend.png`
- `KEN_trend.png`
- `MEX_trend.png`
- `NGA_trend.png`
- `USA_trend.png`
- `ZAF_trend.png`

Duplicate copies inside `CSV_file/` and obsolete rule-of-law image variants are not part of the canonical set.

Color Reference

The following table summarizes the color codes currently used for countries and regional series across the World Bank rule-of-law figures in this repository.

Series	Code	Hex Color	Used In
United States	`USA`	`#1f77b4`	Country trend figures, combined country figure, regional comparison figure
China	`CHN`	`#d62728`	Country trend figures, combined country figure, regional comparison figure
India	`IND`	`#9467bd`	Country trend figures, combined country figure
Brazil	`BRA`	`#2ca02c`	Country trend figures, combined country figure
Mexico	`MEX`	`#e377c2`	Country trend figures, combined country figure
Kenya	`KEN`	`#8c564b`	Country trend figures, combined country figure
Nigeria	`NGA`	`#7f7f7f`	Country trend figures, combined country figure
South Africa	`ZAF`	`#bcbd22`	Country trend figures, combined country figure
Argentina	`ARG`	`#ff7f0e`	Country trend figures, combined country figure
Japan	`JPN`	`#2ca02c`	Regional comparison figure
Africa	`Sub-Saharan Africa`	`#8c564b`	Regional comparison figure
Latin America	`Latin America & Caribbean`	`#ff7f0e`	Regional comparison figure

Series	Code	Hex Color	Used In
Euro area	`Euro area`	`#9467bd`	Regional comparison figure